Process Characterization Report

A-Mab Drug Substance — Ultrafiltration / Diafiltration (Step 10) — PCR-010

Process Development / Manufacturing Science & Technology (MSAT)

2026-09-30

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Approvals

Role	Function	Name (synthetic)	Signature	Date
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Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

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Revision history

Version	Effective date	Description of change
1.0	2026-07-24	Initial issue
2.0	2026-09-30	Reissued under INV-001: re-analysis of the studies, execution dates recorded; changes listed in the document (see INV-001)

Abbreviations

Abbreviation	Meaning
A280	protein concentration by ultraviolet absorbance
AMD	analytical method decision
AMV	analytical method validation
BLA	Biologics License Application
CEX	cation exchange chromatography, as an analytical method
CI	confidence interval
Cpk	process capability index
CPP	critical process parameter
CQA	critical quality attribute
DNA	deoxyribonucleic acid
DS	drug substance
DV	diavolume
GMP	good manufacturing practice
HMW	high molecular weight species
HPLC	high performance liquid chromatography
IAC	intermediate acceptance criterion
ICH	International Council for Harmonisation
icIEF	imaged capillary isoelectric focusing
KPP	key process parameter
LMH	litres per square metre of membrane per hour
Mfg	manufacturing scale
MSAT	Manufacturing Science and Technology
n.a.	not applicable
NOR	normal operating range
nrCE-SDS	non-reduced capillary electrophoresis with sodium dodecyl sulfate
PAR	proven acceptable range
PPQ	process performance qualification
psi	pounds per square inch
QA	Quality Assurance
SD	standard deviation

Abbreviation	Meaning
SDM	scale-down model
SEC	size exclusion chromatography
SOP	standard operating procedure
TFF	tangential flow filtration
TMP	transmembrane pressure
UF/DF	ultrafiltration and diafiltration
UV	ultraviolet

Executive summary

Ultrafiltration and diafiltration (UF/DF) is Step 10, the last step of the A-Mab drug substance process. It concentrates the antibody, exchanges the process buffer for the formulation buffer and delivers the drug substance at its target concentration of 75 g/L. This report gives the results of the characterization of the step under the plan PCP-010 and the master plan PCMP-001. It is part of the Stage 1 record that supports the transfer of the process from the development site in Cambridge, MA to the commercial drug substance site in Grafton, WI.

The step has no designed experiment. The risk assessment RA-001 assigned its three parameters, the number of diavolumes, the transmembrane pressure (TMP) and the final DS concentration, to univariate study. Version 1.0 of this report described the planned univariate study S-UFDF-UNI without a result for any parameter effect. The results reported here come from a further univariate study, S-UFDF-UNI2, made from 2026-09-19 to 2026-09-29, after the QA investigation INV-001 had closed. Each parameter was run at five levels across its range studied, in duplicate, with the other parameters at their set-points. The study ran on a second tangential flow filtration (TFF) skid, EQ-TFF-188. The offsets of the scale-down model from manufacturing scale were measured on the first skid, EQ-TFF-142, and no run compared the two skids. The offsets are applied to the results of the second skid, because the two skids share their design and their membrane and the offsets are small against the criteria (PD-010-01).

Every result met the drug substance criteria for aggregate (≤ 5 % HMW) and for the acidic charge variants (18–40 %), at every level of every parameter. The proven acceptable range (PAR) of each parameter is therefore its range studied, 7–10 DV, 10–15 psi and 65–85 g/L. Aggregate in the final retentate rose with the final DS concentration ($p < 0.001$) and with TMP ($p = 0.015$). Across the ranges studied the two effects amount to 0.14 and 0.06 percentage points, and the highest result lay 4.21 percentage points below the criterion. The A-Mab case study leaves UF/DF out, so no published source gave the size of this effect in advance. No effect of the number of diavolumes on aggregate was detected, and no slope of the acidic charge variants was significant.

PCMP-001 assumed that no step after cation exchange changes aggregate. A check of that assumption on S-UFDF-UNI2 found a rise of aggregate across the NORs of UF/DF of up to 0.104 percentage points, at the one-sided 95 % prediction bound of a single run (PD-010-08). This prediction bound of the aggregate rise across the step tightens the aggregate criterion of the cation exchange pool from 5.00 to 4.90 % HMW.

The residual process buffer fell by a factor of 2.70 for each diavolume, the fall expected when the buffer species pass the membrane freely. At the minimum of 7 DV

it was at most 0.091 % of the process buffer. The buffer-exchange criterion of PCP-010 rests on conductivity readings that are not in the study record, so it could not be judged as planned. The buffer exchange is controlled instead by its minimum number of diavolumes, and the formulation buffer of the drug substance is confirmed against its release limits (PD-010-07).

All three parameters were classified as key process parameters (KPPs), and the step has no critical process parameter. In the integrated process model of 2,000 simulated batches the drug substance has a Cpk of 15.7 for aggregate, 1.05 for the acidic charge variants and 7.3 for residual DNA, which is judged in ng per dose. The Cpk of the acidic charge variants lies between the minimum of 1.00 and the target of 1.33 set by PCMP-001. It is set by the distance from the simulated mean to the lower limit of 18 %. This step can only raise the acidic charge variants, by deamidation, so it moves them away from that limit. Every result of this study lies below the simulated mean. At the mean of the study, with the simulated standard deviation, the Cpk would be 0.63, below the minimum, so this report cannot confirm the simulated capability from its own results (Section 8). Version 1.0 recorded two study events, a TMP excursion to 14.6 psi (DEV-010-01) and a final concentration of 71.5 g/L before a further concentration (DEV-010-02). Neither changes a conclusion. The ranges and classes of this report roll up into the master report PCMR-001.

1 Introduction

1.1 Product and unit operation

A-Mab is a humanized IgG1 monoclonal antibody (CMC Biotech Working Group 2009). Its drug substance is purified by the platform train of Protein A capture, low-pH viral inactivation, cation exchange, anion exchange, small virus retentive filtration and UF/DF. UF/DF is the last step of the train, and its product is the drug substance. The commercial process runs at a production bioreactor working volume of 15,000 L.

The step works across a tangential flow membrane whose pores pass buffer salts and retain the antibody. Ultrafiltration removes permeate and raises the antibody concentration in the retentate. Diafiltration then adds formulation buffer at the rate at which permeate leaves, so that the retentate volume stays constant while the permeable species are washed out. A second concentration brings the retentate to the drug substance target, and a buffer flush recovers it. The step is operated under SOP-2013.

The step is not expected to change the glycan variants of A-Mab, and it is credited with no clearance of host cell protein, DNA or leached Protein A. No viral clearance is claimed for it. It can add to aggregate, which can form at the membrane wall and in the recirculation pump. It can also add to the acidic charge variants, because deamidation continues while the antibody is in solution on the skid (Section 2.1). Aggregate and the acidic charge variants are therefore the attributes measured across the step. The residual process buffer, the step yield and the final concentration are measures of process performance.

Table 1.1 lists the documents to which this report is related. The paired plan is PCP-010, and the parameters were assigned to their studies by the risk assessment RA-001.

Table 1.1: Documents related to this report.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCP-010	Process Characterization Plan (Ultrafiltration / Diafiltration (Step 10))	Sibling — paired document

Document ID	Title	Relationship
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into
INV-001	Investigation Report (Acceptance criteria for proven-acceptable-range analysis set after data review)	Reason for this revision

1.2 Objectives

PCP-010 set the studies the following objectives.

(i) To estimate the offsets of a bench-scale TFF model of the commercial skid from manufacturing scale, as the basis of its qualification (PCMP-001 §7.1).
(ii) To determine the effect of the number of diavolumes, the TMP and the final DS concentration on aggregate and on the acidic charge variants of the drug substance, over the ranges studied.
(iii) To establish the number of diavolumes at which the retentate reaches the formulation buffer, so that the lower edge of the operating range rests on data for A-Mab and not on platform practice alone.
(iv) To establish a PAR for each parameter against the drug substance criteria for aggregate and the acidic charge variants.
(v) To provide the evidence on which the parameters are classified under SOP-4001, and to carry that classification into the control strategy through this report and the master report PCMR-001.

Section 12 states the outcome of each objective.

1.3 Regulatory and scientific basis

The study follows the lifecycle approach to process validation, and this report is part of the record of Stage 1, process design (U.S. Food and Drug Administration 2011). Development follows the enhanced approach of ICH Q8(R2) and ICH Q11, with criticality treated as a continuum and risk managed under ICH Q9 (International Council for Harmonisation 2009, 2012, 2023b). The criticality of the attributes and the logic of parameter classification come from the A-Mab case study (CMC Biotech Working Group 2009). The case study leaves UF/DF out of its characterization (CMC Biotech Working Group 2009, 114), so it gives no expectation and no result for this step. Viral safety rests on the modular and orthogonal clearance of low-pH viral inactivation, anion exchange and small virus retentive filtration (International Council for Harmonisation 2023a). This step contributes no log reduction to it, and the cumulative claim is consolidated in PCMR-001.

1.4 Changes in this revision

Version 1.0 of this report was approved on 2026-07-24. Version 2.0 is issued under the QA investigation INV-001. QA opened it on 2026-08-03, after its review of the master report PCMR-001 Version 1.0 for BLA readiness, and closed it on 2026-09-18. INV-001 found that the criteria used to judge each step in the PAR analysis had been set after the data were seen and appeared in no approved plan (INV-001 finding 1). The master plan PCMP-001 was reissued as Version 2.0 with criteria derived a priori, and PCP-010 was amended to apply it. This report states its results against the amended plan. The changes are the following.

1. Version 1.0 described the planned univariate study S-UFDF-UNI and its outcome, but gave no result for any parameter effect. This version reports S-UFDF-UNI2, a further univariate study made during the re-analysis, and every effect, PAR and class in it rests on that study (Section 4.2). The plan did not foresee the study, and Section 13.2 records it as PD-010-01.
2. The acidic charge variants are judged against both ends of their criterion of 18–40 %. Version 1.0 read them against the upper limit of 40 % alone (INV-001 finding 4), although Version 1.0 of the plan stated the two-sided criterion. The capability of the drug substance is now also assessed against both ends (Section 8).
3. The criteria at this step are the drug substance criteria, which Version 1.0 of the plan also stated. No step follows UF/DF, so PCMP-001 defines no intermediate acceptance criterion (IAC) for it (Section 7.1).
4. The scale-down model is judged by its offsets from manufacturing scale (PCMP-001 §7.1), in place of the tests for a significant difference of Version 1.0. Section 3.1 states the approach this report takes to the qualification decision at a step without a response-surface model.
5. The execution window of each study and the calibration status of the skid it used are given (Section 3.1, Section 4.2). This answers INV-001 finding 5, that the Version 1.0 records gave no execution dates.
6. The events of execution that Version 1.0 recorded as deviations are called study events, because a process characterization study is not GMP manufacture. They keep their numbers DEV-010-01 and DEV-010-02 (Section 13.1).
7. The parameters are classified under the rules of PCMP-001 Version 2.0. An attribute is affected by a parameter when its slope is significant (PCMP-001 §6.2), and a significant effect too small to move the attribute appreciably within the PAR does not by itself make the parameter critical (PCMP-001 §7.7). Section 9 applies both rules.
8. The capability section gives the probability that the drug substance is out of specification, and it judges residual DNA in ng per dose (Section 8).
9. The NOR of the number of diavolumes is a minimum, at least 7 DV. The step counts diavolumes to its set-point and is not run below the minimum (Section 4.1).
10. The buffer exchange is judged on the minimum number of diavolumes and on the release of the drug substance against the limits of its formulation buffer, because the conductivity readings on which PCP-010 judges it are not in the study record

(Section 5.5).

11. The rise of aggregate across the NORs of this step is carried into the criterion of the cation exchange pool, which is tightened from 5.00 to 4.90 % HMW (Section 7.1).

Two statements of Version 1.0 are reversed. Version 1.0 stated that no effect of any parameter on product quality was detected over the ranges assessed. Aggregate rises with the final DS concentration and with TMP (Section 5.3). Version 1.0 also stated that both edges of the diavolume range leave the residual process buffer “far below what the formulation tolerates”. The study record holds no limit for what the formulation tolerates, and this version makes no such claim (Section 5.5).

2 Prior knowledge and quality risk basis

2.1 Platform and prior-product knowledge

The step belongs to the platform train that was used for X-Mab, Y-Mab and Z-Mab (PCMP-001 §2). The places where aggregate can form in the step are the same for every antibody, so that experience applies to the qualitative behaviour of the step. How readily an antibody aggregates depends on its colloidal stability and its self-association, which differ between antibodies. The viscosity of the retentate rises with concentration and differs between antibodies too. For this reason the final DS concentration was studied for A-Mab and was not carried over from the platform. The A-Mab case study leaves the step out (CMC Biotech Working Group 2009, 114). The expectations below therefore come from general knowledge of the tangential flow filtration of antibodies, as PCP-010 §4.1 records them.

Each diavolume removes a fixed fraction of the permeable species that remain in the retentate. PCP-010 §4.1 expects the residual process buffer components to fall "roughly exponentially" with the number of diavolumes. At high antibody concentration two effects can hold the retentate off the composition of the buffer. The antibody carries a net charge, so buffer ions of the opposite charge are enriched in the retentate and ions of the same charge are depleted (the Donnan effect). The antibody also occupies part of the retentate volume, which lowers the concentration of each excipient in the retentate as a whole (volume exclusion). Both effects can move the pH and the excipient levels of the retentate away from the values of the buffer.

The membrane retains the antibody, so the antibody accumulates at the membrane wall in a polarization layer, where its concentration is higher than in the bulk retentate. The retentate also passes many times through the recirculation pump. Aggregate can form at the wall, where the antibody is most concentrated, and in the pump, where cavitation, air-liquid interfaces and particles shed by the pump can start it. Shear alone causes little aggregation of an antibody. Aggregate was therefore expected to rise with the final DS concentration, which raises the concentration in the bulk retentate and, below the limiting flux, at the wall. The limiting flux is the flux beyond which a higher pressure no longer raises the permeate rate. Below it, a higher TMP drives more permeate through the membrane and raises the antibody concentration at the wall. Above it, the concentration at the wall stays at its limiting value. A higher TMP then compacts and thickens the polarization layer, with no gain in flux, so more antibody is held at that concentration. Aggregate was therefore expected to rise with TMP below the limiting flux, and it may still rise above it. It was also expected to rise with the number of passes through the pump, which more diavolumes increase. Within the normal operating ranges (NORs) the exposure is short, and the effects were expected to be small. No source gave their size for A-Mab, so this study measures it

(Section 5.3). The effect of a high final DS concentration and a high TMP on aggregate is this campaign's own addition to the integrated process model (Section 3.5).

Deamidation of asparagine residues continues while the antibody is in solution. It converts a neutral residue into a charged one, so it can only raise the acidic charge variants. The increase grows with the time in solution, at a rate that rises with pH and temperature. The operation lasts longer with more diavolumes, with a higher final concentration, which lowers the flux, and with a lower TMP below the limiting flux. The acidic charge variants were therefore expected to rise, if at all, with the number of diavolumes and the final DS concentration, and to fall with TMP.

2.2 Quality attributes in scope

Table 2.1 lists the attributes in scope at this step, with their drug substance criteria and the criticality assigned with Tool #1 of the A-Mab case study, the product of impact and uncertainty (CMC Biotech Working Group 2009). The studies measured aggregate and the acidic charge variants in the final retentate. Both arrive with the load, and both are formed first in the production bioreactor. Deamidation continues wherever the antibody is held in solution, aggregate also forms at the low-pH elution and hold, and the cation exchange step reduces aggregate (PCMP-001 §5 and Table 7.1). Residual DNA is in scope as an attribute of the drug substance that this step delivers. It is not measured across the step and has no intermediate criterion, so it is judged on the drug substance in the integrated process model (PCMP-001 §7.3, Section 8).

Table 2.1: Quality attributes in scope at the step, with their drug substance criteria and criticality.

CQA	Category	Drug substance criterion	Criticality	Tool #1	At this step
Aggregates (HMW)	size variant	≤ 5 % HMW (SEC)	H	60	Measured in the final retentate by SEC
Acidic charge variants (deamidation)	charge variant	18–40 % (CEX/icIEF)	VL	4	Measured in the final retentate by icIEF
Residual DNA	process impurity	≤ 0.001 ng/dose	VL	6	Not measured. Judged on the drug substance in the integrated process model

H, high criticality. VL, very low criticality. Tool #1, the impact of the attribute on safety and efficacy multiplied by the uncertainty of that impact (A-Mab attribute assessment).

The other attributes of the drug substance register are not measured across the step. Afucosylation, galactosylation and high mannose are formed in the production bioreactor. The membrane retains the antibody whatever its glycoform, so the step is not expected to change their distribution, and they were not measured. Host cell protein and leached Protein A are cleared by the chromatography steps before this one, and PCMP-001 Table 7.1 credits UF/DF with no clearance of either. Fragments form by

proteolytic clipping of the antibody in culture and during the harvest hold, and slowly by hydrolysis of the hinge region in solution. Version 1.0 left fragments out of the studies of this step, and PCP-010 keeps that scope. PCMP-001 Table 7.1 assumes that this step does not change fragments, and the drug substance is tested for fragments at release. Viral clearance is credited to low-pH viral inactivation, anion exchange and virus filtration and is not measured here.

2.3 Risk-based prioritization and parameter selection

RA-001 ranked every process parameter of the drug substance process for its potential impact on the CQAs and on process performance, and for its potential to interact with other parameters (International Council for Harmonisation 2023b). Parameters ranked high, or with a likely interaction, were assigned to multivariate study. Parameters ranked lower, whose range could be supported one factor at a time, were assigned to univariate study (PCMP-001 §4). All three parameters of UF/DF were assigned to univariate study, and no designed experiment was planned for the step.

A higher TMP and a higher final DS concentration both raise the antibody concentration in the polarization layer at the membrane wall, or compact that layer once the flux has reached its limit. A higher final DS concentration raises the viscosity and the bulk concentration of the retentate, and both lower the limiting flux. The flux then reaches its limit at a lower TMP, and above that TMP a higher pressure compacts the layer. The rise of aggregate with TMP may therefore be larger at a high final DS concentration than at a low one. A study that varies one factor at a time cannot resolve such an interaction. It bounds the effect of each parameter at the edge of its range studied, with the others at their set-points (Section 7). PCP-010 §4.3 keeps the rule of Version 1.0 that runs with two parameters at their range edges together are made where a measured attribute approaches its criterion at an edge. Such runs are made before a range is claimed, and they are agreed with Statistics and QA before they are executed.

3 Materials and methods

3.1 Scale-down model and its qualification

The bench-scale model holds the crossflow and the membrane loading at their commercial values, both scaled by membrane area (PCMP-001 §5). The membrane chemistry, the nominal molecular weight cut-off and the feed channel geometry are those of the commercial skid. The TMP, the number of diavolumes and the final concentration do not depend on scale and were set at the values under study. One quantity cannot be matched, the share of the batch held up in the system. With the crossflow and the membrane loading matched per unit area, the antibody passes through the pump as often at bench scale as at commercial scale. The bench skid, however, holds a larger share of its batch in its recirculation loop and has a larger ratio of surface to volume. More antibody is therefore lost to hold-up, and more of it meets surfaces and air-liquid interfaces per gram processed. A slightly lower step yield and slightly more aggregate were expected at bench scale.

The qualification study S-UFDF-SDM ran from 2026-02-25 to 2026-03-06 on the skid EQ-TFF-142, under SOP-1001. It compared three runs of the model at the set-point condition with five batches at manufacturing scale. The outputs compared were the step yield, the aggregate content of the final retentate, the final concentration achieved and the permeate flux at the set-point TMP of 12 psi. Table B.1 gives the individual runs. The scale-down runs were complete on 2026-03-06, before the first characterization run of the step on 2026-03-09, as PCMP-001 §7.1 requires. The qualification record numbers the manufacturing-scale batches without naming them, and it gives no date for the calculation of the offsets. PCP-010 §5.1 asks this report to give both, and Section 13.2 records the gap as PD-010-05.

Under PCMP-001 §7.1 the offset of an output is the mean of the scale-down runs at set-point minus the mean of the manufacturing-scale batches, with its two-sided 90 % confidence interval. Table 3.1 gives the four offsets. The bench skid gave a mean aggregate 0.02 percentage points higher and a mean step yield 0.4 percentage points lower than manufacturing scale. Both differences are in the direction that the larger hold-up predicts. Each interval includes zero, and so do the intervals of the final concentration and the permeate flux. Each interval rests on three runs against five batches and is wide, so this report uses the ends of the interval and not its inclusion of zero.

Table 3.1: Offsets of the scale-down models (scale-down minus manufacturing scale) and the qualification of each model for each output, UF/DF.

Output	Runs		Mean		Offset		Met over NOR (low/est./high)	Qualified
	SDM	Mfg	SDM	Mfg	Estimate	90 % CI		
Aggregates, HMW (%)	3	5	0.71	0.69	0.02	−0.01 to 0.05	no model[a)]	n.a.[a)]
Step yield (%)			96.5	96.9	−0.4	−2.4 to 1.6	no model[a)]	n.a.[a)]
Final concentration (g/L)			75.7	75.0	0.8	−0.9 to 2.4	no model[a)]	n.a.[a)]
Permeate flux (LMH)			37.1	37.3	−0.2	−2.1 to 1.6	no model[a)]	n.a.[a)]

[a)] The step has no response-surface model, and PCMP-001 §7.1 decides the qualification on that model. The offset is recorded, and the report states how it is used.
SDM, scale-down model. Mfg, manufacturing scale. Offset, scale-down mean minus manufacturing mean. Met over NOR: the criterion is met at every combination of NOR low, set-point and NOR high (the NOR in force), with the offset at the low end of its interval, at its estimate (est.) and at the high end.

Of the four outputs only aggregate carries a criterion at this step. The offsets of the step yield, the final concentration and the permeate flux are recorded, and no conclusion depends on them (PCMP-001 §7.1). The qualification compared the aggregate content of the final retentate and not the change in aggregate across the step that PCP-010 Table 5.1 names, because no load was sampled (Section 3.4, PD-010-05). The aggregate offset is therefore subtracted directly from each aggregate result of the final retentate. Every comparison of aggregate with its criterion in this report is stated at manufacturing scale.

Under PCMP-001 §7.1 the decision whether a model is qualified for an output is made on the response-surface model of the step. This step has none. Table 3.1 marks that decision as not applicable, and PCP-010 §7.1 leaves the decision to this report (PD-010-06). This report keeps the logic of PCMP-001 §7.1, that a model is qualified for an output when the conclusion over the NOR is the same with the offset at its estimate and at either end of its interval. Without a model, the conclusion over the NOR is whether every individual aggregate result at a level inside the NOR meets the criterion at manufacturing scale. All 24 aggregate results inside the NOR met the criterion in all three cases, so the model is qualified for aggregate under this approach. The highest of them was 0.82 % HMW, with the offset at the end of its interval that is least favourable to the criterion.

The univariate study S-UFDF-UNI2 ran on the second skid EQ-TFF-188. The qualification measured the offsets on EQ-TFF-142 only, and no run compared the two skids. The offsets of EQ-TFF-142 are applied to the results of EQ-TFF-188 (PD-010-01). This is accepted on two grounds. The two skids share their design and their membrane, so the hold-up and the surface that the antibody meets per gram, which cause the offsets, are expected to be alike. The offsets are also small against the criteria. The aggregate offset is 0.02 percentage points, and the highest aggregate result of S-UFDF-UNI2

lies 4.21 percentage points below its criterion (Section 7.2). As measured, the six set-point runs of S-UFDF-UNI2 gave a mean aggregate of 0.74 %, against 0.71 % for the scale-down runs of S-UFDF-SDM and 0.69 % for the manufacturing-scale batches.

The acidic charge variants and the residual process buffer were not compared at both scales. Their offsets are unknown, not zero, and both are judged as measured (PCMP-001 §7.1). For the acidic charge variants this rests on the kinetics of deamidation, which depend on the pH and the temperature of the solution and on the time spent in it, and not on the scale of the skid. At the same TMP and crossflow the flux is the same at both scales, so a run of the model lasts as long as the commercial step at the same settings (PCP-010 §5.1). For the buffer exchange it rests on the passage of the buffer species through the membrane. That passage is a property of the membrane and of the species, so the fraction removed in each diavolume does not depend on scale.

3.2 Operation

The step receives the filtrate pool of the small virus retentive filtration step. Under SOP-2013 the pool is concentrated by ultrafiltration, diafiltered at constant volume against formulation buffer for the number of diavolumes of the run, concentrated a second time to the final target and recovered with a buffer flush. At the set-point condition the step runs 8 DV at a TMP of 12 psi to a final concentration of 75 g/L. The composition of the formulation buffer is fixed by the drug product program and was a constant of the studies. Under PCP-010 §5.2 every run is loaded from a single pool of the virus filtration step, so that the variability of the load does not confound the comparison between levels. Section 12 of the plan gives a bridging rule if one pool is not sufficient. The study record of S-UFDF-UNI2 does not identify the load pool of its runs, so it cannot show that one pool served the whole study (PD-010-04).

3.3 Analytical methods

The procedures and validated methods are listed in Table 3.2. Aggregate is measured by size exclusion chromatography (SEC) under AMV-3011 and reported as the percentage of high molecular weight species (HMW). The acidic charge variants are measured by imaged capillary isoelectric focusing (icIEF) under AMV-3013 and reported as the percentage of acidic species. Protein concentration is measured by ultraviolet absorbance (A280) under AMV-3019, and the final concentration and the step yield rest on that measurement. PCP-010 follows the buffer exchange by the conductivity and the pH of the retentate, which are in-process readings of the skid instrumentation and not validated assays. The study record of S-UFDF-UNI2 gives one result of the buffer exchange per run, the residual process buffer. This is the part of the process buffer, the buffer in which the antibody enters the step, that is left in the final retentate, given as a percentage. The record names neither the buffer species measured nor the method, and it holds no conductivity or pH reading (PD-010-02).

Table 3.2: Procedures and method validations referenced in this document.

Type	Reference	Title
Standard operating procedure	SOP-1001	Qualification of Scale-Down Models
	SOP-2013	Operation of the Ultrafiltration / Diafiltration (Formulation) Step
	SOP-4001	Process-Parameter Classification and Control-Strategy Definition
Analytical method validation	AMV-3011	Size-Variants (SEC-HPLC)
	AMV-3019	Protein Concentration by A280 (UV)
	AMV-3013	Charge Variants (icIEF)

The studies of this step do not measure fragments, a scope that PCP-010 keeps from Version 1.0. The analytical method decision AMD-001 of 2026-03-20 set non-reduced CE-SDS (nrCE-SDS) as the method for fragments at every step, so it changes no result of this report. AMD-001 moved fragments away from SEC because host cell protein that elutes in the low molecular weight window of SEC is counted as fragment. No such interference applies to aggregate, and SEC remains its method (PCMP-001 §7.8). Aggregate was measured by SEC in the qualification study S-UFDF-SDM and in the univariate study S-UFDF-UNI2 (PCP-010 §5.3).

The aggregate content of A-Mab drug substance is low, and a change across this step may be of the size of the repeatability of SEC. PCP-010 §5.3 keeps the rules of Version 1.0 for such small changes. A measured change is judged against the pure error of the study and not against the nominal precision of the method. Noise near the low end of the method range is attributed to the assay unless the change measured across the range is larger than the pure error. Section 3.5 states how this rule and the slope test were applied together. PCP-010 §5.3 also asks for the variability of each method over the range of the samples. The study record reports none, so the pure error of the duplicate runs is the only measure of variability used here (PD-010-04).

3.4 Sampling plan

Table 3.3 compares the samples that PCP-010 Table 5.2 planned with the results that the record of S-UFDF-UNI2 holds. The criteria for aggregate and the acidic charge variants apply to the final retentate, and the record holds both results for every run, together with the final concentration and the step yield. It holds one result of the residual process buffer per run and no conductivity or pH reading (PD-010-02). The load pool, the pre-diafiltration retentate and the permeate were not sampled (PD-010-03). Without a load result the change in aggregate and in the acidic charge variants across the step cannot be calculated. The study therefore measures how the parameters change the final retentate, and not what the step adds to its load. Without a permeate result the retention of the antibody by the membrane is not confirmed either.

Table 3.3: Samples planned in PCP-010 and results held in the record of S-UFDF-UNI2.

Sample	Point in the step	Tests planned in PCP-010	Results in the record of S-UFDF-UNI2
Load pool	Virus filtration filtrate, before ultrafiltration	A280, SEC, icIEF	None
Pre-diafiltration retentate	End of the first concentration	A280	None
Diafiltration series	After each diavolume, in the runs of the diavolume series	Conductivity, pH	One residual process buffer result per run. No conductivity or pH reading
Final retentate	After the final concentration and the recovery flush	A280, SEC, icIEF	Final concentration, aggregate and acidic charge variants in every run, with the step yield from the mass balance
Permeate	Composite over the run	A280	None

Conductivity and pH are in-process readings of the skid instrumentation, not validated assays (PCP-010 §5.3).

3.5 Statistical methods

For each parameter and each output, the individual results were regressed on the setting of the parameter over the range studied, and the slope was tested against zero at $\alpha = 0.05$ (PCP-010 §5.5, PCMP-001 §6.2). Each series has 10 runs, so each slope has 8 degrees of freedom. An attribute is called affected by a parameter when its slope is significant at that level. The change across the range studied is the slope times the width of the range. It is reported with its 95 % confidence interval beside the p-value, so that statistical and practical significance are read apart. The criterion, and not the p-value, decides whether a range is acceptable. No response-surface model was fitted, because the study has no factorial structure and no interaction can be estimated from it. The model-adequacy criteria of PCMP-001 §7.2 apply to response-surface models and not to these slopes. The one fit of several parameters in this report is the main-effects fit of aggregate used for the verification of Section 7.1. It is judged on the first three criteria of PCMP-001 §7.2, a significant model, no significant lack of fit and the limit on overfitting. PCMP-001 §7.5 names these three for a model whose bound widens a criterion, and this report applies them to the fit whose bound tightens one.

The duplicate runs at each level give the pure error of the study, the combined variability of the process and the assay, as a pooled standard deviation with 15 degrees of freedom for each output. PCP-010 §5.5 takes the pure error from the replicated set-point condition. This study replicates every level, so the pure error is pooled over all of them (PD-010-06). PCP-010 §5.3 leaves to this report how its rule for small changes and the slope test are applied together. This report applies both to aggregate, the attribute whose changes the rule concerns. The slope test decides whether aggregate is affected, and the change across the range is compared with the pure error. For aggregate the two agreed for every parameter (Section 5.3). The acidic

charge variants lie near 21 %, against a limit of quantitation of 1 % for icIEF. The rule for the low end of the method range therefore does not apply to them, and the slope test alone decides.

The buffer exchange was analysed by fitting the exponential decay that the mechanism predicts (Section 2.1), as the natural logarithm of the residual process buffer regressed on the number of diavolumes. PCP-010 §5.5 plans that fit on the conductivity series of each run. The study record holds no such series, so the fit was made across the runs of the diavolume series, on one result per run (PD-010-06). Complete passage of the buffer species through the membrane gives a slope of −1 on that scale.

Each attribute result was judged on its own against its criterion, as PCMP-001 §6.2 requires for a quality attribute and PCP-010 §7.2 states for this step. Aggregate was judged at manufacturing scale and the acidic charge variants as measured (Section 3.1). The PAR of a parameter is the contiguous span of levels over which every attribute measured on the final retentate met its criterion, with the other parameters at their set-points (PCP-010 §8).

The study record of S-UFDF-UNI2 does not give the order in which its runs were made (PD-010-04). It cannot show the randomized order that PCP-010 §6.2 requires, and a drift over the course of the study could not be separated from a slope.

The capability of the drug substance comes from the integrated process model of PCMP-001 §6.5. The model runs 2,000 simulated batches through every step. Each process parameter is drawn from a normal distribution centred on its set-point, with a standard deviation of one sixth of the width of its NOR, and the draw is clipped to the range studied. The NOR of the number of diavolumes is a minimum and has no upper edge. The model draws it around its set-point of 8 DV with a standard deviation of 0.33 DV, one sixth of a range of 7–9 DV, and never below the minimum of 7 DV or above 10 DV. The model includes the effects of the UF/DF parameters on aggregate and no effect of them on the acidic charge variants. PCMP-001 §7.9 sets a target Cpk of 1.33 and a minimum of 1.00.

4 Study design

4.1 Factors, ranges and the knowledge space

Table 4.1 gives the parameters with their set-points, NORs, ranges studied and the levels of S-UFDF-UNI2. Each range studied extends beyond the NOR at one end or at both, and it is the knowledge space within which a PAR can be claimed.

Table 4.1: Parameters of the step: set-points, normal operating ranges, ranges studied and the levels of the univariate study.

Parameter	Unit	Set-point	NOR	Range studied	Levels of S-UFDF-UNI2
Number of diavolumes	DV	8	≥ 7[a)]	7–10	7, 7.5, 8, 9, 10
Transmembrane pressure	psi	12	10–14	10–15	10, 11, 12, 13.5, 15
Final DS concentration	g/L	75	70–80	65–85	65, 70, 75, 80, 85

[a)] A minimum. The parameter is counted to its set-point and is not run below this value.

The number of diavolumes was studied over 7–10 DV. Its NOR is a minimum of 7 DV. The step counts diavolumes to its set-point of 8 DV and is not run below the minimum. Each further diavolume lowers the residual process buffer further. It also adds passes through the pump, which could raise aggregate, and time on the skid, which could raise the acidic charge variants. The range studied therefore extends to 10 DV to measure those effects. Across 7–10 DV the slope of aggregate was not significant ($p = 0.894$), and the 95 % confidence interval of the change runs from -0.06 to 0.07 percentage points (Section 5.3). The study does not go below the minimum. TMP was studied over 10–15 psi against a NOR of 10–14 psi. The range extends above the NOR to measure whether a higher TMP, which raises the antibody concentration at the membrane wall or compacts the polarization layer, raises aggregate. Its lower edge is the lower edge of the NOR. A lower TMP lowers the flux and lengthens the operation, and the study does not characterize that side. The final DS concentration was studied over 65–85 g/L against a NOR of 70–80 g/L, on both sides. Above the target a higher final concentration raises the viscosity and the bulk concentration of the retentate, which lowers the flux and holds more of the antibody at high concentration. The range extends below the NOR as well, so that a retentate concentrated short of the target is also covered. No PAR is claimed beyond the range studied (PCMP-001 §7.6).

4.2 Univariate assessment

PCP-010 §6.2 planned the univariate study S-UFDF-UNI. Each parameter was to be run at the two edges of its range studied and at its set-point, with the other two at their set-points. The set-point condition was shared by the series and replicated, and the order of execution was to be randomized across the study. The planned study ran from 2026-03-09 to 2026-03-27 on EQ-TFF-142, the skid of the qualification. No result of it for any parameter effect is in the record, and Version 1.0 described its outcome without one.

The further univariate study S-UFDF-UNI2 was made from 2026-09-19 to 2026-09-29, during the re-analysis and after INV-001 closed on 2026-09-18. It gave the parameters the recorded results that Version 1.0 lacked. Each parameter was run at five levels, in duplicate, with the other two at their set-points. The low and high levels are the edges of the range studied, and the two intermediate levels lie halfway between the set-point and each edge (Table 4.1). The study has 30 runs, six of them at the set-point condition. It ran on EQ-TFF-188, a second UF/DF TFF skid, which was in calibration at execution, with its next calibration due on 2027-03-31. The qualification and the planned univariate study on the first skid both closed before its next calibration, which was due on 2026-04-22.

S-UFDF-UNI2 is not a designed experiment, and no run combines the levels of two parameters. The effects, PARs and classes of this report rest on one parameter at a time. Only the verification of aggregate in Section 7.1 fits the three parameters together, with no interaction term (PD-010-08). The study differs from the plan in its levels, its replicates and its skid (PD-010-01).

5 Results

5.1 Reproducibility of the duplicate runs

Table 5.1 gives the pure error of each output and the results of the six set-point runs, two in each series. The pure error of aggregate was 0.023 percentage points and that of the acidic charge variants 0.46 percentage points. The pure error of the step yield was 2.08 percentage points, and that of the final concentration 1.13 g/L. The set-point runs of the three series gave a mean aggregate of 0.74 % with a standard deviation of 0.018 percentage points.

Table 5.1: Pure error of each output, and mean, SD and extremes of the 6 set-point runs, S-UFDF-UNI2.

Output	Pure error	Mean	SD	Lowest	Highest
Aggregates, HMW (%)	0.023	0.74	0.018	0.71	0.76
Acidic variants (%)	0.46	20.6	0.55	19.9	21.4
Residual process buffer (%)	0.0025	0.0333	0.0022	0.0306	0.0362
Step yield (%)	2.08	95.6	0.66	94.4	96.2
Final concentration (g/L)	1.13	75.4	1.21	73.5	76.6

Pure error, the pooled standard deviation of the duplicate runs at each level, with 15 degrees of freedom. Set-point runs, the 6 runs at the set-point condition, 2 in each series. Aggregate as measured.

5.2 Effects of the parameters

Table 5.2 gives, for each parameter and output, the change across the range studied that the slope predicts, its 95 % confidence interval and the p-value of the slope. Of the 15 slopes, four were significant at $\alpha = 0.05$. The number of diavolumes affected the residual process buffer. TMP and the final DS concentration affected aggregate, and the final DS concentration affected the final concentration achieved, which is its own setting. No slope of the acidic charge variants or of the step yield was significant, and neither TMP nor the final DS concentration had a significant slope on the residual process buffer.

Table 5.2: Change of each output across the range studied, from the slope of the individual results on the setting, S-UFDF-UNI2.

Parameter and range studied	Output (unit of the change)	Change across the range studied	95 % confidence interval	p
Number of diavolumes, 7–10 DV	Aggregates, HMW (pp)	0.00	−0.06 to 0.07	0.894
	Acidic variants (pp)	−0.5	−1.4 to 0.4	0.243
	Residual process buffer (pp)	−0.0777	−0.1020 to −0.0534	< 0.001[a)]
	Step yield (pp)	−1.7	−4.6 to 1.1	0.198
	Final concentration (g/L)	−0.5	−2.0 to 0.9	0.406
Transmembrane pressure, 10–15 psi	Aggregates, HMW (pp)	0.06	0.01 to 0.10	0.015[a)]
	Acidic variants (pp)	0.4	−0.2 to 1.0	0.190
	Residual process buffer (pp)	−0.0010	−0.0057 to 0.0037	0.631
	Step yield (pp)	−1.6	−5.0 to 1.9	0.324
	Final concentration (g/L)	0.2	−2.7 to 3.1	0.888
Final DS concentration, 65–85 g/L	Aggregates, HMW (pp)	0.14	0.09 to 0.18	< 0.001[a)]
	Acidic variants (pp)	−0.2	−1.2 to 0.8	0.628
	Residual process buffer (pp)	−0.0017	−0.0058 to 0.0024	0.374
	Step yield (pp)	−1.9	−6.4 to 2.6	0.356
	Final concentration (g/L)	20.0	18.4 to 21.5	< 0.001[a)]

[a)] Slope significant at α = 0.05. pp, percentage points. Change across the range studied, the slope times the width of the range. Each slope rests on 10 runs, 8 degrees of freedom. A change of aggregate is taken to be the same at both scales, because the aggregate offset was measured at the set-point only and is assumed constant across settings.

5.3 Aggregate

Every aggregate result met the criterion of ≤ 5 % HMW at manufacturing scale, also with the offset at the end of its interval that is least favourable to the criterion. The highest result was 0.79 % HMW, at 80 g/L in the final DS concentration series, and it lay 4.21 percentage points below the criterion. Figure 5.1 shows the individual results against each parameter.

Aggregate rose with the final DS concentration. Across the range studied of 65–85 g/L the slope predicts a rise of 0.14 percentage points (95 % confidence interval 0.09 to 0.18, p < 0.001), and across the NOR a rise of 0.07. Aggregate also rose with TMP, by 0.06 percentage points across 10–15 psi (95 % confidence interval 0.01 to 0.10, p = 0.015) and by 0.04 across the NOR. Both changes across the range are larger than the pure error of 0.023 percentage points. Under the rule of PCP-010 §5.3, a change larger than the pure error is not assay noise, so neither is attributed to the assay. Both agree in direction with the expectation of Section 2.1. The larger change, for the final DS concentration, is 3 % of the margin between the highest result and the criterion. This report judges both changes small against the criterion (PD-010-06).

No effect of the number of diavolumes on aggregate was detected (p = 0.894). The 95 % confidence interval of the change across 7–10 DV runs from −0.06 to 0.07 percentage points. More diavolumes add passes through the pump, and Section 2.1 expected aggregate to rise with them. The data do not show that rise at the precision of the study, and an effect within the interval cannot be excluded.

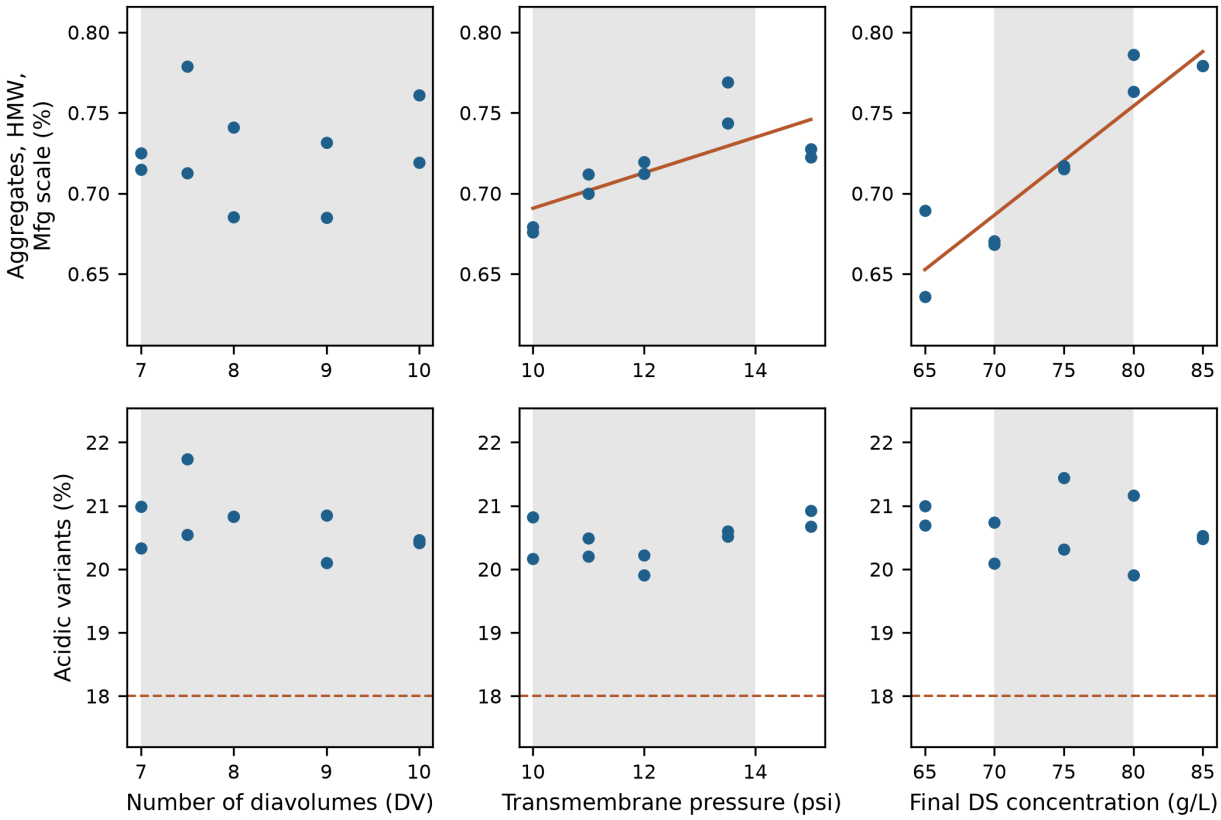


Figure 5.1: Aggregate at manufacturing scale (upper row) and acidic charge variants as measured (lower row) in the final retentate against each parameter, S-UFDF-UNI2. Each point is one run. Aggregate at manufacturing scale is the result minus the aggregate offset. The grey band is the NOR, open to the right for the number of diavolumes, whose NOR is a minimum. A solid line is the fitted slope where the slope is significant. The dashed line is the lower limit of the criterion for the acidic charge variants. The aggregate criterion lies far above the plotted range.

5.4 Acidic charge variants

Every result of the acidic charge variants lay within the criterion of 18–40 %, at both ends. The lowest result, 19.9 %, was measured twice, at the set-point condition of the transmembrane pressure series and at 80 g/L in the final DS concentration series. It lay 1.9 percentage points above the lower limit. The highest was 21.7 %, 18.3 percentage points below the upper limit. No slope was significant ($p = 0.243$, $p = 0.190$ and $p = 0.628$ for the number of diavolumes, TMP and the final DS concentration). The 95 % confidence intervals of the change across the range studied run from −1.4 to 0.4, from −0.2 to 1.0 and from −1.2 to 0.8 percentage points, in the same order. A change of up to 1.4 percentage points across a range therefore cannot be excluded. Section 2.1 expected deamidation to raise the acidic charge variants with the number

of diavolumes and the final DS concentration, which lengthen the operation, and to lower them with TMP, which shortens it.

The acidic charge variants enter the step with the load from the virus filtration step. They are formed first in the production bioreactor, whose report PCR-003 characterizes them, and deamidation adds to them wherever the antibody is held in solution. Without a load result the study cannot say how much the step adds at the set-point condition (Section 3.4). Deamidation can only raise the acidic charge variants, so whatever the step adds moves them toward the upper limit of 40 % and away from the lower limit of 18 %.

5.5 Buffer exchange

The residual process buffer fell exponentially with the number of diavolumes (Figure 5.2). The slope of its natural logarithm on the number of diavolumes was −0.995 (95 % confidence interval −1.033 to −0.957, $R^2 = 0.998$). The interval includes −1, the slope that complete passage of the buffer species through the membrane gives. Each diavolume removed 63 % of the process buffer that remained, and the residual fell by a factor of 2.70 per diavolume. The two runs at the set-point of 8 DV in the diavolume series gave a mean residual of 0.034 % of the process buffer. At the minimum of 7 DV the two results were 0.083 and 0.091 %. At the top of the range studied, 10 DV, neither result exceeded 0.0046 %. The residual process buffer had no significant slope on TMP ($p = 0.631$) or on the final DS concentration ($p = 0.374$), as a wash-out that depends on the number of diavolumes alone would predict.

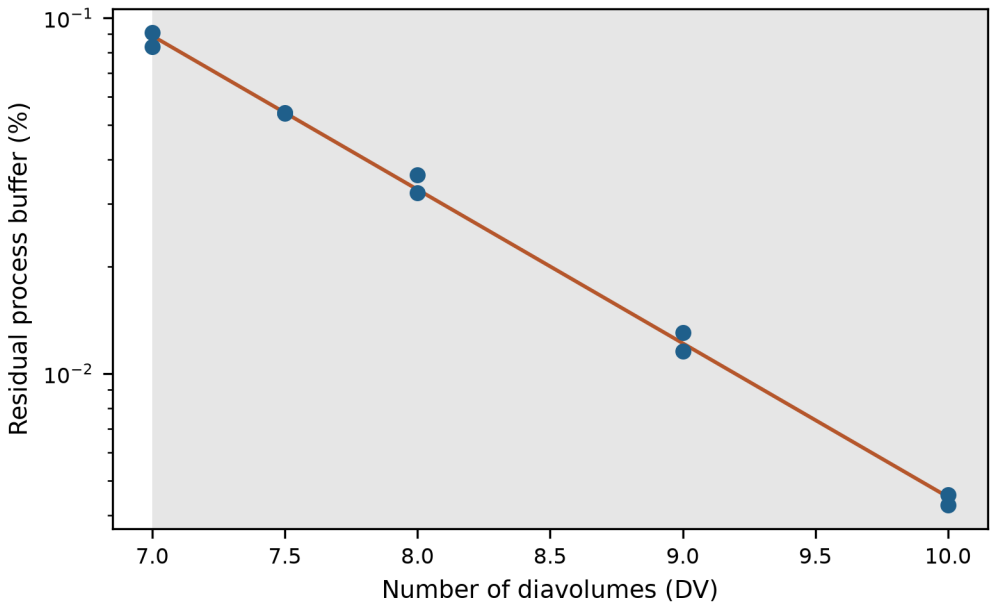


Figure 5.2: Residual process buffer in the final retentate against the number of diavolumes, S-UFDF-UNI2, on a logarithmic scale. Each point is one run. The line is the exponential fall fitted to the runs of the series. On this scale an exponential fall is a straight line, and a levelling off would show as a bend. The grey band is the NOR, a minimum, open to the right.

The two results at each level of the diavolume series lay wholly above both results at the next level, at every step from 7 DV to 10 DV.

PCP-010 judges the buffer exchange on the conductivity of the retentate. Section 5.5 of the plan places the end of the exchange at the number of diavolumes at which "the conductivity of the retentate has reached the plateau set by the formulation buffer and no further change is measured over the following diavolume". Section 7.2 accepts the buffer exchange when the conductivity and the pH have reached that plateau at or below 7 DV. The study record holds no conductivity or pH reading (PD-010-02), so the criterion could not be judged as planned. The residual process buffer cannot stand in for the conductivity. It falls by a constant fraction with each diavolume, so it does not level off within any range that an assay can resolve.

The buffer exchange is controlled instead by its minimum number of diavolumes, 7 DV, the lower edge of a one-sided NOR. The formulation buffer of the drug substance is confirmed against its release limits (PD-010-07). The result of this study is the residual process buffer at each number of diavolumes, at most 0.091 % of the process buffer at the minimum.

Objective (iii), to establish the number of diavolumes at which the retentate reaches the formulation buffer, is not met. The study gives the residual process buffer after each number of diavolumes, but not that number.

5.6 Step yield and final concentration

The step yield ranged from 91.0 to 99.2 %, with a mean of 95.7 %. No slope of the step yield was significant. The 95 % confidence intervals of its change across the range studied run from −4.6 to 1.1 percentage points for the number of diavolumes, from −5.0 to 1.9 for TMP and from −6.4 to 2.6 for the final DS concentration. The step yield has no criterion and is reported for information.

The final concentration followed its setting in the series of the final DS concentration, with a slope of 0.998 (95 % confidence interval 0.920 to 1.075). Each result lay within 1.2 g/L of its setting. In the other two series, run at the target of 75 g/L, the final concentration ranged from 73.3 to 76.6 g/L, inside the NOR of 70–80 g/L. The pure error of the final concentration, 1.13 g/L, compares with a precision of the A280 method of 1.2 %, which is 0.9 g/L at the target.

5.7 Mechanistic interpretation

The rise of aggregate with the final DS concentration and with TMP is consistent with aggregate forming in the polarization layer at the membrane wall. The final DS concentration raises the bulk concentration at the end of the operation, and below the limiting flux it raises the concentration at the wall too. Below the limiting flux TMP raises the concentration at the wall by driving more permeate through the membrane. Above it, the concentration at the wall stays at its limiting value, and a higher TMP compacts and thickens the layer, so more antibody is held at that concentration. The study did not measure the flux across the TMP range, so it cannot say on which side of the limiting flux the set-point lies. Within the NOR the two effects amount to 0.07 and 0.04 percentage points.

No effect of the number of diavolumes on aggregate was detected, although more diavolumes add passes through the pump. Diafiltration is made before the second concentration, at a lower antibody concentration than the final one. If aggregate forms mainly where the concentration at the wall is highest, extra diavolumes at that lower concentration would add little. The study did not measure where in the step aggregate forms, so this explanation is not tested.

The acidic charge variants did not change detectably over the ranges studied, although each parameter changes the time the antibody spends in solution. The study record does not give the duration of the runs, so the study cannot relate the acidic charge variants to the time in solution. Any deamidation in the step would raise them, toward the upper limit of the criterion.

The fall of the residual process buffer by a factor of 2.70 for each diavolume is consistent with free passage of the buffer species through the membrane, which diafiltration at constant volume relies on. The study did not measure the pH or the excipient levels of the final retentate. It therefore gives no evidence on the Donnan and volume-exclusion effects that can move them away from the values of the buffer at high concentration.

6 Design space

No design space is defined for UF/DF. A design space is a multivariate region (International Council for Harmonisation 2009). A study that varies one parameter at a time cannot define one, because it gives no result where two parameters are away from their set-points at the same time. The step is described instead by the univariate PAR of each parameter, which equals its range studied (Section 7). These data support the movement of one parameter within its PAR, with the others at their set-points. Section 7.3 gives only the size that an interaction would need to bring aggregate to its criterion. An excursion outside a NOR in manufacture is dispositioned against the PAR (PCP-010 §8). A planned move of an operating range is a change, and it is assessed under change management before it is made (International Council for Harmonisation 2008).

7 Proven acceptable ranges

7.1 Acceptance basis

No step follows UF/DF, so PCMP-001 Table 7.1 defines no IAC for it, and the drug substance criteria apply directly (PCMP-001 §7.6). Aggregate must not exceed 5 % HMW, and the acidic charge variants must lie between 18 and 40 %. Both criteria come from the drug substance CQA register (PCMP-001 §3), and no result of these studies enters them.

The a priori derivation of the criteria upstream of this step assumes that no step after cation exchange changes aggregate, and that no step after low-pH viral inactivation changes the acidic charge variants (PCMP-001 §7.3). On that assumption the cation exchange pool carried the drug substance criterion for aggregate itself, 5.00 % HMW. This study shows that two parameters of UF/DF raise aggregate within their NORs. On the sum of the two slopes the rise at the corner of 80 g/L and 14 psi is 0.06 percentage points above the level at the set-point.

This report checks the assumption of no aggregate change after cation exchange on the results of S-UFDF-UNI2, as PCMP-001 Version 2.0 §7.5 states. PCP-010 §7.2 planned no verification at this step (PD-010-08). The aggregate results, with the aggregate offset subtracted, are fitted to the main effects of the three parameters, with all runs pooled and no interaction term. The fit is evaluated at every combination of NOR low, set-point and NOR high, as for a step with a response-surface model. For the number of diavolumes, whose NOR is a minimum, the upper value of the combinations is 9 DV, the value that also sets the spread of the simulation (Section 3.5). At the least favourable combination the fit predicts a mean rise of 0.057 percentage points above its value at the set-point. The one-sided 95 % prediction bound of a single run puts the rise at 0.104 percentage points. The fit meets the first three criteria of PCMP-001 §7.2. The model is significant ($p < 0.001$), it has no significant lack of fit ($p = 0.147$), and R^2 exceeds predicted R^2 by 0.15, within the limit of 0.2. PCMP-001 §7.5 rule 1 states that where the worst case predicted within the NOR is worse than the platform worst case, the prediction bound of a single batch replaces the platform value. Here the bound replaces the assumption of no change, and every aggregate criterion upstream carries it. The criterion of the cation exchange pool is tightened from 5.00 to 4.90 % HMW, and the cation exchange report PCR-007 judges its pool against that value. The fit assumes that the parameters act independently, and Section 7.3 gives the size that an interaction would need to matter.

For the acidic charge variants no slope was significant. The change across the step was not measured, because no load was sampled (PD-010-03), and a rise of up to 1.4 percentage points across a range cannot be excluded (Section 5.4). The assumption of

PCMP-001 §7.3 that no step after low-pH viral inactivation changes the acidic charge variants ignores the deamidation on the UF/DF skid, which raises them (Section 2.1). Deamidation moves the acidic charge variants away from the lower limit of 18 %, which is the limiting side (Section 8), so the assumption errs on the safe side for that limit. Toward the upper limit of 40 %, every result of this study lay at least 18.3 percentage points inside.

7.2 Results against the criteria

Table 7.1 gives, for each level of each parameter, the lowest and highest aggregate result at manufacturing scale and the lowest and highest result of the acidic charge variants. PCP-010 §7.3 accepts a level when every attribute measured on the final retentate of its runs meets its criterion. Every attribute result of every level met its criterion, so every level is acceptable. The plan passes a parameter over its range when every level is acceptable and the change measured across the range is small against the criterion. It leaves the threshold for a small change to this report (PD-010-06). This report gives the changes (Table 5.2) and the margins (Table 7.1) and judges each change small against its criterion. The largest significant change of an attribute across a range, 0.14 percentage points of aggregate for the final DS concentration, is 2.7 % of the criterion of 5 % HMW. The acidic charge variants have no significant change. A change of up to 1.4 percentage points across a range cannot be excluded, and that is smaller than the margin of 1.9 percentage points between the lowest result and the lower limit. No level failed, so the rules for a failing level were not used, and the study record shows no repeated run.

Table 7.1: Lowest and highest result of each level, S-UFDF-UNI2. Every result met its criterion.

Parameter	Level	Setting	Inside the NOR	Aggregates, HMW, Mfg scale (%)	Acidic variants (%)
Number of diavolumes (DV)	Low	7	Yes	0.71–0.72	20.3–21.0
	Low-mid	7.5	Yes	0.71–0.78	20.5–21.7
	Set-point	8	Yes	0.69–0.74	20.8
	Mid-high	9	Yes	0.69–0.73	20.1–20.8
	High	10	Yes	0.72–0.76	20.4–20.5
Transmembrane pressure (psi)	Low	10	Yes	0.68	20.2–20.8
	Low-mid	11	Yes	0.70–0.71	20.2–20.5
	Set-point	12	Yes	0.71–0.72	19.9–20.2
	Mid-high	13.5	Yes	0.74–0.77	20.5–20.6
	High	15	No	0.72–0.73	20.7–20.9
Final DS concentration (g/L)	Low	65	No	0.64–0.69	20.7–21.0
	Low-mid	70	Yes	0.67	20.1–20.7
	Set-point	75	Yes	0.71–0.72	20.3–21.4
	Mid-high	80	Yes	0.76–0.79	19.9–21.2
	High	85	No	0.78	20.5

The criteria are ≤ 5 % HMW for aggregate and 18–40 % for the acidic variants. Aggregate at manufacturing scale, the scale-down result minus the aggregate offset. Acidic variants as measured. Each level has 2 runs, and a single value means that both gave it at the precision shown. The NOR of the number of diavolumes is a minimum.

7.3 Proven acceptable ranges of the parameters

Table 7.2 gives the PAR of each parameter. Each PAR rests on aggregate and the acidic charge variants, the attributes the study measured, and on no other attribute. For the final DS concentration the PAR extends 5 g/L beyond the NOR at each end. For TMP it extends 1 psi above the NOR, and its lower edge is the lower edge of the NOR. The NOR of the number of diavolumes is a minimum of 7 DV, and the PAR starts at that minimum. The PAR ends at 10 DV, because the study gives no result above it.

Table 7.2: Proven acceptable ranges of the parameters, with the other parameters at their set-points.

Parameter	NOR	PAR	PAR beyond the NOR, low	PAR beyond the NOR, high
Number of diavolumes	≥ 7 DV[a)]	7–10 DV	0 DV	n.a.
Transmembrane pressure	10–14 psi	10–15 psi	0 psi	1 psi
Final DS concentration	70–80 g/L	65–85 g/L	5 g/L	5 g/L

[a)] A minimum. The parameter is counted to its set-point and is not run below this value. Each PAR is the range studied. It rests on aggregate and the acidic charge variants, the attributes measured in the final retentate.

Three limitations apply to these PARs, as PCP-010 §8 states them.

1. Each PAR holds with the other parameters at their set-points. It does not describe the step when two parameters sit at their range edges at the same time.
2. PCMP-001 §6.4 defines a second analysis, which predicts each attribute from a fitted model while the other parameters are drawn at random around their set-points. This step has no such model, so that analysis was not performed. For a parameter studied one factor at a time the range rests on its own data instead (PCMP-001 §4).
3. A PAR cannot be wider than the range studied. Each PAR here is the range studied, and no claim is made beyond it (PCMP-001 §7.6).

PCP-010 §4.3 asks for runs with two parameters at their range edges together where a measured attribute approaches its criterion at an edge. It leaves the threshold for “approaches” to this report (PD-010-06). At every edge the highest aggregate result lay at least 4.22 percentage points below its criterion, and the acidic charge variants lay at least 2.2 percentage points inside theirs. This report judges that no attribute approached its criterion at an edge, and no such runs were made.

The interaction of TMP and the final DS concentration that Section 2.3 describes was not measured. Its possible size can be set against the criterion for a single run. The highest set-point run gave 0.74 % HMW at manufacturing scale. At the corner of the ranges studied, 85 g/L and 15 psi, the two slopes add 0.10 percentage points to give 0.84 % HMW. An interaction would have to add a further 4.16 percentage points there, about 41 times the rise that the two slopes give, to bring that run to the criterion. The study cannot exclude an interaction, but one of that size is far outside the effects it measured.

8 Process capability and robustness

UF/DF delivers the drug substance, so this report gives the capability of the drug substance for the attributes that the step can change and for residual DNA. Table 8.1 gives the result of the integrated process model for 2,000 simulated batches, with each process parameter drawn as Section 3.5 describes (PCMP-001 §6.5). The model includes an effect of the final DS concentration and of TMP on aggregate. The table gives each Cpk, the count of simulated batches outside the criterion and the tail probability of a normal distribution with the simulated mean and standard deviation. PCMP-001 §7.9 sets a target Cpk of 1.33 and a minimum of 1.00.

Table 8.1: Capability of the drug substance in the integrated process model.

CQA	Criterion	Mean	SD	Cpk	Simulated batches outside, of 2,000	Normal tail, per million batches
Aggregates (HMW)	≤ 5 % HMW (SEC)	0.78	0.090	15.66	0	< 0.001
Acidic charge variants (deamidation)	18–40 % (CEX/icIEF)	22.4	1.39	1.05	1	821
Residual DNA	≤ 0.001 ng/dose	0.000170	0.0000379	7.30	0	< 0.001

Mean and SD of the simulated batches. Residual DNA in ng per dose, for a dose of 700 mg of antibody. Normal tail, the probability beyond the criterion of a normal distribution with the simulated mean and SD, two-sided for a two-sided criterion.

Aggregate has a Cpk of 15.7, above the target, and no simulated batch lay outside its criterion. The mean simulated aggregate is 0.78 % HMW, with a standard deviation of 0.090 percentage points. A Cpk this large says only that the criterion lies far beyond the simulated batches. Aggregate is therefore judged on the individual results of the study (Section 7.2) and on the count of simulated batches outside the criterion. With none of 2,000 outside, the one-sided 95 % upper confidence bound of the probability of a batch outside is 0.15 %. That bound covers the sampling of the simulation and not the uncertainty of the integrated process model.

The acidic charge variants have a Cpk of 1.05 in the integrated process model, between the minimum and the target. The Cpk is set by the distance from the simulated mean to the lower limit of the criterion, 18 %. Of the 2,000 simulated batches, one fell outside the criterion, and the normal tail probability is 821 per million. This step does not move the acidic charge variants toward that lower limit. Deamidation can only raise them, and the integrated process model gives the UF/DF parameters no effect on them. The attribute is formed in the production bioreactor, and its report PCR-003 gives the parameters that drive it and the controls that address them.

The results of this study do not confirm the simulated level. All 30 results of S-UFDF-UNI2 lay between 19.9 and 21.7 %, below the simulated mean of 22.4 %. Their level is the level of the load, which was not sampled (PD-010-03). At the mean of the study, 20.6 %, with the simulated standard deviation, the Cpk would be 0.63, below the minimum. The capability of the acidic charge variants therefore depends on the level that the upstream steps deliver, and this report cannot settle it from its own results. PCMR-001 assesses it across the whole process.

Residual DNA is judged in ng per dose, the unit of its criterion of 0.001 ng per dose. That criterion is the acceptable range that the A-Mab case study claims for residual DNA (CMC Biotech Working Group 2009, Table 2.17, pp. 47–48). The dose is A-Mab's 10 mg/kg (CMC Biotech Working Group 2009, 26) for a body weight of 70 kg. The body weight is an assumption of this campaign, and with it a dose carries 700 mg of antibody. The simulated drug substance carries a mean of 0.00017 ng per dose, or 0.00024 pg per mg of antibody, with a Cpk of 7.3. No simulated batch lay outside the criterion, and the same upper bound of 0.15 % applies. This level is not a measured result. It is computed from the DNA clearance that the integrated process model credits to each purification step, which is an input from outside the data of this step. It lies about 4,100 times below the limit of quantitation of the qPCR method for residual DNA, 1 pg per mg (AMV-3014), so that method could not quantify it. The reports of the capture and polishing steps (PCR-005, PCR-007 and PCR-008) give the DNA results of their studies. Residual DNA has no IAC, because the platform history gives no worst-case DNA clearance for the steps after Protein A, so no criterion can be carried back (PCMP-001 §7.3). Platform experience shows that DNA is consistently cleared from the process (CMC Biotech Working Group 2009, 47). In the testing plan of the A-Mab case study an attribute of low criticality with moderate to high process capability would likely not require testing (CMC Biotech Working Group 2009, 57). No DNA clearance is credited to UF/DF.

9 Parameter classification

The parameters were classified under SOP-4001 by the rules of PCMP-001 §7.7, which follow the decision logic of the A-Mab case study (CMC Biotech Working Group 2009). A critical process parameter (CPP) has "an effect on a CQA that is significant and practically relevant within the characterized range, and a high risk of falling outside its acceptable range in operation". A well-controlled critical process parameter has the same kind of effect but a low risk, because it is readily controlled and its PAR is wide against its NOR. A KPP "affects process performance, such as yield or concentration, and has no effect on a CQA within its range". PCMP-001 §7.7 adds that "a significant effect that is too small to move the attribute appreciably within the PAR does not by itself make a parameter critical". A significant effect on a CQA therefore makes a parameter critical unless the effect is too small to be practically relevant, and PCMP-001 and SOP-4001 set no threshold for that. Each class below is stated with the size of each effect across the PAR and the margin of the PAR to the NOR (PCP-010 §7.4). Table 9.1 summarises the evidence.

Table 9.1: Evidence for the classification of the parameters.

Parameter	Class	Change of aggregate across the PAR (pp)	p, aggregate	p, acidic variants	Process performance output	PAR beyond the NOR, low / high
Number of diavolumes	KPP	0.00 (−0.06 to 0.07)	0.894	0.243	Residual process buffer, measured	0 DV / n.a.
Transmembrane pressure	KPP	0.06 (0.01 to 0.10)	0.015	0.190	Permeate flux, from the mechanism. Not measured across the range	0 psi / 1 psi
Final DS concentration	KPP	0.14 (0.09 to 0.18)	< 0.001	0.628	Final concentration, measured	5 g/L / 5 g/L

pp, percentage points of HMW. Change, the slope times the width of the PAR, with its 95 % confidence interval in brackets. p, of the slope of the attribute on the parameter. n.a., not applicable, because the NOR of the number of diavolumes is a minimum.

- The number of diavolumes was classified as a KPP. More diavolumes lower the residual process buffer, by a factor of 2.70 for each diavolume (Section 5.5). Its slopes on aggregate (p = 0.894) and on the acidic charge variants (p = 0.243) were not significant, so no effect on either attribute was detected within the range studied, at the precision of the duplicates. Its NOR is a minimum of 7 DV, and its PAR runs from that minimum to 10 DV.
- TMP was classified as a KPP. It has a significant effect on aggregate, a CQA (p = 0.015). Across its NOR of 10–14 psi the slope raises aggregate by 0.04 percentage points, and across its PAR of 10–15 psi by 0.06. The highest aggregate result of the study lay 4.21 percentage points below the criterion. This report judges

an effect of that size too small to be practically relevant, and the KPP class of both TMP and the final DS concentration rests on that judgement. The effect is carried into product quality all the same, because the prediction bound of 0.104 percentage points that sets the criterion of the cation exchange pool includes it (Section 7.1). The effect of TMP on process performance, through the permeate flux and so the duration of the operation, was not measured across the TMP range (Table 9.1). The flux was measured only at the set-point, in S-UFDF-SDM. The PAR extends 1 psi above the NOR and not below it.

- The final DS concentration was classified as a KPP. It sets the concentration of the drug substance, which followed it with a slope of 0.998 (Section 5.6). It has a significant effect on aggregate, a CQA ($p < 0.001$). Across its NOR of 70–80 g/L the slope raises aggregate by 0.07 percentage points, and across its PAR of 65–85 g/L by 0.14, against the same margin of 4.21 percentage points. The bound of 0.104 percentage points behind the criterion of the cation exchange pool includes this effect too. The PAR extends 5 g/L beyond the NOR at each end.

The step has no CPP and no well-controlled CPP. Its three parameters are KPPs, which are controlled within their NORs so that the step performs consistently (CMC Biotech Working Group 2009).

10 Contribution to the control strategy

The control strategy for the drug substance process is consolidated in PCMR-001. This report contributes the following.

1. The NORs of the parameters are ≥ 7 DV, 10–14 psi and 70–80 g/L. The parameters are KPPs, and their PARs are the ranges studied. The NOR of TMP and of the final DS concentration lies inside its PAR. The NOR of the number of diavolumes is a minimum, and the PAR covers counts up to 10 DV. The parameters are controlled within their NORs so that the step delivers the drug substance consistently at its target concentration, with the residual process buffer that Section 5.5 gives.
2. A higher final DS concentration and a higher TMP raise aggregate, through their action on the polarization layer at the membrane wall. Within their NORs the rise is at most 0.104 percentage points at the prediction bound of a single run, and the criterion of the cation exchange pool carries that bound (Section 7.1). In the studies TMP was monitored in process and the final concentration was checked by A280, and these two checks detected the study events DEV-010-01 and DEV-010-02 (Section 13.1).
3. The buffer exchange is controlled by its minimum number of diavolumes, 7 DV. At the minimum the residual process buffer was at most 0.091 % of the process buffer. The formulation buffer of the drug substance is confirmed against its release limits (PD-010-07).
4. The membrane chemistry and the membrane supplier are part of the qualified state of the step, and a change to either requires a new risk assessment before these data are applied (PCP-010 §12). The composition of the formulation buffer is fixed by the drug product program, and a change to it would change the target of the buffer exchange.

Other attributes of the drug substance are controlled at other steps. The glycan variants and the acidic charge variants are formed in the production bioreactor (PCR-003). Fragments form by proteolytic clipping in culture and during the harvest hold (PCR-003, PCR-004), and slowly by hydrolysis of the hinge region in solution, and they are tested at release. Host cell protein and residual DNA are cleared by the chromatography steps (PCR-005, PCR-007 and PCR-008). Leached Protein A enters the process at capture (PCR-005) and is cleared by the polishing steps. Viral clearance is credited to low-pH viral inactivation, anion exchange and virus filtration (PCR-006, PCR-008 and PCR-009), and the cumulative claim is consolidated in PCMR-001.

11 Discussion

Two parameters of UF/DF, the final DS concentration and TMP, raise aggregate. Both rises exceed the pure error of the study, and this report judges them small against the criterion. They still matter upstream. PCMP-001 had assumed that no step after cation exchange changes aggregate, and the bound for a single run of 0.104 percentage points now tightens the criterion of the cation exchange pool to 4.90 % HMW. More diavolumes lower the residual process buffer, by the factor that free passage of the buffer species predicts. No slope of the acidic charge variants was significant. Every result met the drug substance criteria, and each parameter has a PAR equal to its range studied.

The scale-down model is qualified for aggregate under the approach of Section 3.1, which judges the aggregate results with the offset at its estimate and at either end of its interval. Three points limit that confidence. The offsets were measured on the first skid, and the study ran on the second. The discrepancy entry for the use of the offsets of the first skid on the second (PD-010-01) accepts it because the skids share their design and their membrane and the offsets are small against the criteria. The acidic charge variants and the residual process buffer were not compared at both scales, so their representativeness rests on the kinetics of deamidation and on the passage of the buffer species through the membrane. The qualification record does not name the manufacturing-scale batches (PD-010-05). The highest aggregate result lies far below its criterion (Section 7.2), so none of these points changes a conclusion about aggregate.

The study record of S-UFDF-UNI2 is incomplete against the plan in ways that bear on its reading. No load was sampled, so the change across the step is unknown and the study measures only how the parameters change the final retentate (PD-010-03). No conductivity or pH was read, so the buffer-exchange criterion could not be judged as planned (PD-010-02), and the buffer exchange rests on the minimum number of diavolumes and on the release of the drug substance (PD-010-07). The order of the runs and the load pool are not recorded, so a drift over the study cannot be separated from a slope (PD-010-04).

The main scientific limitation is the univariate design, which cannot measure an interaction. TMP and the final DS concentration both act on the polarization layer at the membrane wall, so an interaction between them would raise aggregate most where both are high. The prediction bound of the aggregate rise in Section 7.1 assumes no interaction, and Section 7.3 sets the size that one would need against the criterion. Each slope rests on 8 degrees of freedom, so an effect smaller than the confidence intervals of Table 5.2 may have gone undetected, for the acidic charge variants in particular.

The capability of the acidic charge variants rests on the level that the upstream steps deliver. Every result of this study lies below the mean of the integrated process model, and Section 8 shows that the Cpk at the level of the study would fall below the minimum. PCMR-001 assesses this attribute across the whole process.

The receiving site will run the process performance qualification (PPQ) batches of the transfer protocol PTP-001 against the ranges and classes of this report (PCMP-001 §2).

12 Conclusions

1. The bench-scale TFF model is qualified for aggregate under the approach of Section 3.1. Its offsets from manufacturing scale were applied to the results of the second skid. The discrepancy entry for that use (PD-010-01) accepts it because the skids share their design and their membrane and the offsets are small against the criteria (objective i).
2. Aggregate rises with the final DS concentration and with TMP, by 0.14 and 0.06 percentage points across the ranges studied. No effect of the number of diavolumes on aggregate was detected, and no slope of the acidic charge variants was significant (objective ii). The rise across the NORs tightens the criterion of the cation exchange pool to 4.90 % HMW.
3. The residual process buffer falls by a factor of 2.70 for each diavolume and is at most 0.091 % of the process buffer at the minimum of 7 DV. The buffer-exchange criterion of PCP-010 could not be judged as planned. The buffer exchange is controlled by the minimum number of diavolumes and by the release of the drug substance against the limits of its formulation buffer. Objective (iii) is not met, because the number of diavolumes at which the retentate reaches the formulation buffer was not established.
4. Every result met the drug substance criteria for aggregate and the acidic charge variants. The PARs are 7–10 DV, 10–15 psi and 65–85 g/L, each equal to its range studied (objective iv).
5. The three parameters are KPPs, and the step has no CPP. TMP and the final DS concentration raise aggregate by 0.04 and 0.07 percentage points across their NORs, against a margin of 4.21 percentage points to the criterion, and their KPP class rests on the size of these effects. The drug substance has a Cpk of 15.7 for aggregate, 1.05 for the acidic charge variants and 7.3 for residual DNA in the integrated process model. The Cpk of the acidic charge variants lies between the minimum and the target of PCMP-001. At the mean of this study it would be 0.63, so it depends on the level that the upstream steps deliver (objective v).
6. Neither study event changes a conclusion. The departures from the plan are recorded as eight discrepancies (Section 13.2). The results roll up into PCMR-001.

13 Study events and discrepancies from the plan

A process characterization study is not GMP manufacture and raises no deviation. PCP-010 §13 keeps the wording of Version 1.0, which records a departure from the plan during execution as a deviation. This report records two kinds of entry instead. The study events are the events of execution that Version 1.0 recorded, under the numbers that Version 1.0 gave them. The discrepancies from the plan are the departures from PCP-010, both work that the plan did not foresee and requirements of the plan that were not met.

13.1 Study events

Version 1.0 recorded two study events at this step, which Table 13.1 lists.

Table 13.1: Study events recorded at the step.

Event	Summary	Detected during	Outcome
DEV-010-01	Transmembrane-pressure excursion above NOR during concentration	in-process TMP monitoring	run retained in the analysis
DEV-010-02	Final DS concentration below target on one run; topped up	final concentration check	run retained in the analysis

The record of each event names neither the study nor the run in which it occurred. DEV-010-01 names the skid EQ-TFF-142, the first skid, on which the qualification S-UFDF-SDM and the planned univariate study S-UFDF-UNI ran. No result of the planned study is in the record, so an event in one of its runs affects no result of this report. The assessments below cover the case in which an event occurred in a run of the qualification.

13.1.1 DEV-010-01, transmembrane pressure above the NOR

During concentration the TMP of one run rose to 14.6 psi, above the upper edge of its NOR of 14 psi. In-process TMP monitoring detected the excursion. The investigation attributed it to a transient fouling of the membrane, and the run was retained. The

excursion lay within the range studied of 10–15 psi, over which every aggregate result of S-UFDF-UNI2 met its criterion (Section 7.2). Suppose the run was one of the scale-down runs of S-UFDF-SDM and was held at that TMP throughout. The upper end of the 95 % confidence interval of the slope of aggregate on TMP then puts its effect at no more than 0.050 percentage points. It would have raised the mean of the three scale-down runs, and with it the aggregate offset, by at most 0.017 percentage points. With the offset lowered by that amount, every aggregate result at manufacturing scale would still meet the criterion.

13.1.2 DEV-010-02, final concentration below target

The final concentration check of one run measured 71.5 g/L by A280 (AMV-3019), below the target of 75 g/L and inside the NOR of 70–80 g/L. The investigation attributed the shortfall to the residual hold-up volume of the skid, the property of the bench skid that Section 3.1 expects to lower its step yield. The record does not say how the hold-up lowered the final concentration. The retentate was concentrated further to the target, which kept it in the pump loop for a further period at the highest concentration of the run. The run was retained. The record gives neither the concentration after the correction nor the run.

The correction could have raised the aggregate of that run. Had the run been one of the scale-down runs of S-UFDF-SDM, it would enter both the offset of the final concentration and the aggregate offset. The final concentration has no criterion at this step. The aggregate offset of 0.020 percentage points lies between 0.010 and 0.026 when any one scale-down run is left out. With the smallest of these, the highest aggregate result at manufacturing scale is 0.80 % HMW, 4.20 percentage points below the criterion, so no conclusion depends on the run.

13.2 Discrepancies from the plan

Table 13.2 lists the eight discrepancies from PCP-010.

Table 13.2: Discrepancies from the plan PCP-010.

Discrepancy	Subject	Planned	Done	Reason
PD-010-01	Univariate study with recorded results, made during the Version 2.0 re-analysis	PCP-010: a univariate study of each UF/DF parameter at its range edges and set-point (study S-UFDF-UNI), which Version 1.0 described and narrated without a result for any parameter effect	a further univariate study, each parameter at five levels in duplicate with the others at set-point (study S-UFDF-UNI2), run on the second skid EQ-TFF-188, which the qualification did not compare with EQ-TFF-142; the offsets of EQ-TFF-142 are applied to its results, accepted because the skids share their design and membrane and the offsets are small against the criteria	to give the UF/DF parameters the recorded results that Version 1.0 lacked
PD-010-02	No conductivity or pH readings during diafiltration	PCP-010: conductivity and pH of the retentate after each diavolume, to judge when the retentate reaches the formulation buffer (PCP-010 Table 5.2, §5.5)	one result of residual process buffer per run, with no conductivity series and no pH; the buffer-exchange criterion cannot be judged as the plan states it	the in-process readings of the diafiltration are not in the study record

Discrepancy	Subject	Planned	Done	Reason
PD-010-03	Load, pre-diafiltration and permeate not sampled	PCP-010: the load pool, the retentate before diafiltration and the permeate sampled, so that the change across the step and the retention of the antibody are known (PCP-010 §4.2, §5.4)	the final retentate only; the change across the step is not known	the samples were not scheduled in the study record
PD-010-04	Records of execution not kept	PCP-010: runs in random order from one identified pool of the virus filtration step, method variability reported over the sample range, samples assayed within their stability period, and a second-person review of every calculation (PCP-010 §5.2, §5.3, §5.4, §6.2, §9, §12)	neither the run order nor the load pool is in the record, no method variability is reported, and the stability period and the review are not recorded	the record gives no reason

Discrepancy	Subject	Planned	Done	Reason
PD-010-05	Qualification record incomplete	PCP-010: the manufacturing-scale batches named, the offsets estimated before the first characteriza-tion run, and the change of aggregate across the step compared at both scales (PCMP-001 §7.1, PCP-010 §5.1, Table 5.1)	the batches are numbered without names, the record gives no date for the offsets, and the aggregate of the final retentate was compared	the record gives no reason
PD-010-06	Rules the report states where the plan leaves them open, and the pure error	PCP-010: PCP-010 leaves the decision rules of §4.3, §5.3, §7.1 to §7.3 and Table 5.1 to the report, and takes the pure error from the three set-point replicates (§5.5)	the report states and applies each rule, fits the buffer exchange across the runs of the diavolume series, and pools the pure error from the duplicates at every level	the plan delegates the rules, and the record holds one buffer-exchange result per run

Discrepancy	Subject	Planned	Done	Reason
PD-010-07	Buffer exchange judged on the minimum diavolumes and drug substance release	PCP-010: buffer exchange complete when conductivity has reached the plateau set by the formulation buffer and no further change is measured over the following diavolume (PCP-010 §5.5)	the criterion could not be judged as planned, because the record holds no conductivity reading (PD-010-02); the buffer exchange is controlled by its minimum number of diavolumes, the lower edge of a one-sided NOR, and the formulation buffer of the drug substance is confirmed against its release limits	the number of diavolumes is a minimum, because each further diavolume lowers the residual process buffer further, and the drug substance is released against the limits of its formulation buffer

Discrepancy	Subject	Planned	Done	Reason
PD-010-08	Aggregate verified at UF/DF	PCP-010: no verification at this step (PCP-010 §7.2); PCMP-001 Version 1.0 assumed that no step after cation exchange changes aggregate	the increase of aggregate across the step was verified on study S-UFDF-UNI2 by a main-effects fit across all its runs, evaluated at the least favourable NOR combination, as PCMP-001 Version 2.0 §7.5 states; its one-sided 95 % single-run bound replaced the assumption of no increase and tightened the cation exchange pool criterion	the study measured a rise of aggregate within the NOR, which the assumption of no change after cation exchange does not cover

The first, PD-010-01, is work that the plan did not foresee. PCP-010 planned the univariate study S-UFDF-UNI, which ran but left no result for any parameter effect. The further study S-UFDF-UNI2 was made during the re-analysis to give the parameters recorded results, and every result, PAR and class of this report rests on it (Section 4.2). It ran on the second skid, with the offsets of the first applied to its results (Section 3.1). No result for a parameter effect existed when it was made, so no conclusion rests on runs added after a result was known.

PD-010-02 and PD-010-07 concern the buffer exchange. Without the conductivity and pH readings the criterion of PCP-010 could not be judged as planned, and the buffer exchange is controlled by the minimum number of diavolumes and the release of the drug substance (Section 5.5).

The absence of load, pre-diafiltration and permeate samples (PD-010-03) leaves the change across the step unknown for aggregate and the acidic charge variants, and the retention of the antibody unconfirmed (Section 3.4). The missing execution records

(PD-010-04) mean that the randomized order, the single load pool, the method variability, the stability period of the samples and the second-person review of the calculations cannot be shown from the record (Section 3.2, Section 3.5). The incomplete qualification record (PD-010-05) names no batches and gives no date for the offsets, and the final retentate was compared in place of the change across the step (Section 3.1).

PD-010-06 covers the rules that PCP-010 leaves to this report. They are the qualification decision without a response-surface model (Section 3.1), the use of the rule for small changes beside the slope test (Section 3.5), the threshold for a small change and for an attribute that approaches its criterion (Section 7.2, Section 7.3), the fit of the buffer exchange across runs and the pure error pooled from every level (Section 3.5).

PD-010-08 is the verification of aggregate at this step, which PCP-010 §7.2 did not plan. PCMP-001 Version 1.0 assumed that no step after cation exchange changes aggregate, and PCMP-001 Version 2.0 §7.5 checks that assumption on S-UFDF-UNI2. A fit of the main effects of the three parameters across all runs of the study, evaluated at the least favourable NOR combination, gave a one-sided 95 % prediction bound of a single run of 0.104 percentage points. That bound replaced the assumption of no change and tightened the criterion of the cation exchange pool to 4.90 % HMW (Section 7.1).

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Appendix A. Results of the univariate study

Table A.1 lists every result of S-UFDF-UNI2, by parameter and setting. Aggregate is given as measured and at manufacturing scale, the result minus the aggregate offset of Table 3.1.

Table A.1: Results of the univariate study S-UFDF-UNI2, by parameter and setting.

Parameter	Setting	Run	Aggregates, HMW (%)	Aggregates, Mfg (%)	Acidic variants (%)	Residual buffer (%)	Step yield (%)	Final conc. (g/L)
Number of diavolumes (DV)	7	1	0.75	0.72	21.0	0.0909	96.2	75.2
		2	0.73	0.71	20.3	0.0828	99.1	75.0
	7.5	1	0.80	0.78	21.7	0.0540	94.5	75.1
		2	0.73	0.71	20.5	0.0538	96.6	74.8
	8	1	0.71	0.69	20.8	0.0322	96.2	75.7
		2	0.76	0.74	20.8	0.0362	96.0	76.6
	9	1	0.75	0.73	20.8	0.0116	97.2	74.3
		2	0.71	0.69	20.1	0.0130	97.0	75.1
	10	1	0.78	0.76	20.4	0.0043	93.8	75.4
		2	0.74	0.72	20.5	0.0046	95.4	74.1
Transmembrane pressure (psi)	10	1	0.70	0.68	20.8	0.0330	95.4	75.5
		2	0.70	0.68	20.2	0.0357	99.2	73.5
	11	1	0.73	0.71	20.5	0.0323	97.5	73.3
		2	0.72	0.70	20.2	0.0306	94.1	76.6
	12	1	0.73	0.71	19.9	0.0340	95.4	73.5
		2	0.74	0.72	20.2	0.0306	95.5	76.5
	13.5	1	0.76	0.74	20.5	0.0368	93.3	74.7
		2	0.79	0.77	20.6	0.0323	96.8	73.7
	15	1	0.74	0.72	20.7	0.0327	95.9	76.2
		2	0.75	0.73	20.9	0.0301	95.2	74.1
Final DS concentration (g/L)	65	1	0.66	0.64	20.7	0.0312	94.8	64.8
		2	0.71	0.69	21.0	0.0362	96.7	65.1
	70	1	0.69	0.67	20.1	0.0340	98.0	70.9
		2	0.69	0.67	20.7	0.0336	93.3	70.3
	75	1	0.74	0.71	21.4	0.0317	95.8	75.6
		2	0.74	0.72	20.3	0.0352	94.4	74.4
	80	1	0.78	0.76	19.9	0.0329	92.9	80.7
		2	0.81	0.79	21.2	0.0359	96.9	80.9
	85	1	0.80	0.78	20.5	0.0307	91.0	83.8
		2	0.80	0.78	20.5	0.0319	96.4	85.8

Run, the replicate at the setting. The other parameters at their set-points. Aggregates, HMW, as measured. Aggregates, Mfg, at manufacturing scale. Residual buffer, the residual process buffer as a percentage of the process buffer. Final conc., the final concentration.

Appendix B. Runs of the scale-down qualification

Table B.1 lists the runs of S-UFDF-SDM behind the offsets of Table 3.1.

Table B.1: Runs of the scale-down qualification S-UFDF-SDM, at the set-point condition.

Scale	Run	Step yield (%)	Aggregates, HMW (%)	Final conc. (g/L)	Permeate flux (LMH)
Scale-down model	1	96.1	0.70	75.5	36.0
	2	95.0	0.70	75.5	38.0
	3	98.5	0.73	76.1	37.3
Manufacturing scale	1	96.0	0.70	72.8	37.0
	2	96.7	0.66	76.2	38.3
	3	95.8	0.73	76.2	35.1
	4	98.5	0.68	74.5	37.4
	5	97.6	0.69	75.0	38.8

Final conc., the final concentration achieved. Permeate flux at the set-point TMP. The record does not name the manufacturing-scale batches.

Appendix C. Analytical methods summary

Table C.1 summarises the validated methods of the studies of this step.

Table C.1: Validated analytical methods of the studies of the step.

Output	Technique	Method validation	Title	Reported as	Sample
Aggregates (HMW)	Size exclusion chromatography	AMV-3011	Size-Variants (SEC-HPLC)	% HMW	Final retentate
Acidic charge variants	Imaged capillary isoelectric focusing	AMV-3013	Charge Variants (icIEF)	% acidic species	Final retentate
Protein concentration	Ultraviolet absorbance (A280)	AMV-3019	Protein Concentration by A280 (UV)	g/L	Final retentate, and the mass balance of the step yield

The buffer exchange is followed by in-process readings of the skid instrumentation, which are not validated assays (PCP-010 §5.3).